CS 300 – Chapter 13 Lectures

Additional Material

Jose de Lima

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Key idea Chapter map

Where this sits in CS 300. Chapter 13 is the *comparative-sorts encyclopedia + list-idioms chapter*: not one algorithm or one structure, but the survey of *every sort the reader is likely to encounter in production code* plus the canonical algorithmic idioms over linked lists. ch. 3 covered the foundational comparison sorts (insertion / selection / merge); ch. 7 covered heapsort. This chapter closes the loop by (a) deriving the $\Omega(n \log n)$ comparison-sort lower bound that motivates *why* those sorts cluster at $n \log n$ and *why* the linear-time sorts (counting / radix / bucket) need extra structure, (b) walking the production hybrid sorts (`std::sort` = introsort, Python / Java `sort` = Timsort, Java primitive `sort` = dual-pivot quicksort, Rust `sort_unstable` = pdqsort), and (c) collecting the list idioms (cycle detection, in-place reversal, sorted merge, Nth-from-end, intersection) that fall out of singly- and doubly-linked structure. ch. 4 introduced linked lists as a structure; this chapter is where the algorithms *on* linked lists live.

What you add to your toolkit:

- **Decision-tree lower bound.** Any comparison-based sort runs in $\Omega(n \log n)$ time in the worst case. Decision-tree model + height-of-tree-with-$n!$-leaves argument. Explains why merge / heap / introsort all cluster at $n \log n$ and not below.
- **Linear-time non-comparison sorts.** Counting sort ($O(n + k)$ for keys in $[0, k)$), radix sort ($O(d(n + b))$ for d-digit keys in base b), bucket sort ($O(n)$ expected for uniform keys). Each beats the $n \log n$ lower bound by exploiting key structure instead of pairwise comparison.
- **Hybrid sorts.** Introsort (quicksort + heapsort fallback + insertion sort for small ranges — Musser 1997, behind `std::sort`). Timsort (merge sort exploiting natural runs — Tim Peters 2002, behind Python `list.sort` and Java `Arrays.sort` for objects). Dual-pivot quicksort (Yaroslavskiy 2009, behind Java's primitive `Arrays.sort`). Pdqsort (pattern-defeating quicksort — Orson Peters 2016, behind Rust `sort_unstable`). Why these beat any single algorithm in practice.
- **Quickselect and median-of-medians.** `std::nth_element`'s algorithmic core. Average $O(n)$ via quickselect with random pivot; worst-case $O(n)$ via median-of-medians (groups-of-five recursion).
- **List-specific algorithmic idioms.** Floyd's tortoise-and-hare cycle detection ($O(1)$ space cycle finding + cycle-length + cycle-start), in-place list reversal (iterative + recursive), sorted-merge of two sorted lists, Nth-from-end (two-pointer), list intersection.
- **Sort-algorithm decision table.** How to pick the right sort given (a) input size, (b) data distribution (random / nearly-sorted / bounded-integer / string), (c) stability requirement, (d) memory budget, (e) parallelism requirement.

Mastery checklist — you should be able to, without references:

1. State the *decision-tree lower bound* $\Omega(n \log n)$ for any comparison-based sort and sketch the argument: a comparison sort's execution is a binary decision tree, must have at least $n!$ leaves (one per permutation), so its height (= worst-case number of comparisons) is at least $\lceil \log_2(n!) \rceil = \Omega(n \log n)$.
2. Implement quicksort with both *Lomuto* and *Hoare* partition. Identify which one `std::sort` (introsort) uses and why (median-of-three pivot + Hoare-style two-pointer scan in libstdc++).
3. State counting sort and argue why it is $O(n + k)$ and stable. Identify when counting sort outperforms quicksort (when $k = O(n)$, i.e. key universe linear in input size).
4. State radix sort (LSD form) and argue why it is $O(d(n + b))$ for d-digit keys in base b. Identify the base-vs-passes tradeoff (more bits per pass ⇒ fewer passes but bigger counting-sort buckets; production radix sorts often use $b = 256$, one byte per pass).
5. Pick the right sort at first sight: tiny array (insertion sort, $n \leq 20$); in-place fast general-purpose (introsort = `std::sort`); stable (Timsort or merge sort); bounded integer keys (counting or radix); nearly-sorted input (Timsort or insertion); must-not-allocate (introsort, since merge sort needs $O(n)$ scratch). Justify each.
6. Implement Floyd's tortoise-and-hare cycle detection on a singly-linked list. State the cycle-length and start-of-cycle algorithms (cycle length: keep one pointer fixed, advance the other until they meet again; start of cycle: from meeting point, move one pointer back to head and advance both at speed 1 until they meet).
7. State the introsort algorithm (quicksort + heapsort fallback when recursion depth exceeds $2 \log_2 n$ + insertion sort for small ranges, typically 16–32 elements in production C++ STL). Identify the recursion-depth threshold and small-array threshold.

Looking ahead. Past the cs-300 boundary, sorting keeps extending into regimes this chapter does not touch: external sorting (when the data does not fit in RAM — merge-from-disk k-way merges, the engine behind classic `sort(1)` on multi-gigabyte inputs); GPU sorts (radix sort is the standard parallel sort on GPUs — CUB, Thrust); concurrent / lock-free sorts (parallel samplesort, ips4o); approximate / probabilistic sorts (Spreadsort variants for partially-sorted streams). The production-references notebox in §13.11 is the entry point.

Key idea

Chapter 13 covers a lot of ground but the unifying lens is *which sort fits which problem.* Every section either introduces a new sort, derives a bound that constrains which sorts are possible, or builds out the linked-list algorithms whose analysis lives at the same complexity-class level as the sorts. The closing decision table in §13.9 is the pay-off: by the end of the chapter, every cell in that table should read like a sentence you already know.

1 13.0 The comparison-sort lower bound

Before walking specific sorts, we derive the bound that constrains every comparison-based sort: no comparison sort can do better than $\Omega(n \log n)$ comparisons in the worst case. This is *not* a property of any one algorithm — it is a property of the comparison-sort *model*, and it is what motivates the non-comparison sorts (counting / radix / bucket) in §13.6 and §13.7.

The decision-tree model

Definition —

Decision-tree model of a sort. Model an arbitrary comparison-based sort as a binary tree. Each *internal* node is a comparison "is $a_i \leq a_j$?" between two input positions; the two children correspond to the yes / no outcomes. Each *leaf* is a complete permutation of the input that the algorithm outputs. The sort's worst-case number of comparisons on n elements equals the height of this tree.

The decision-tree model captures *any* comparison sort: insertion, selection, merge, quicksort, heapsort, Timsort, introsort, pdqsort. Each algorithm corresponds to a different shape of tree, but every algorithm in the family is one tree.

The bound

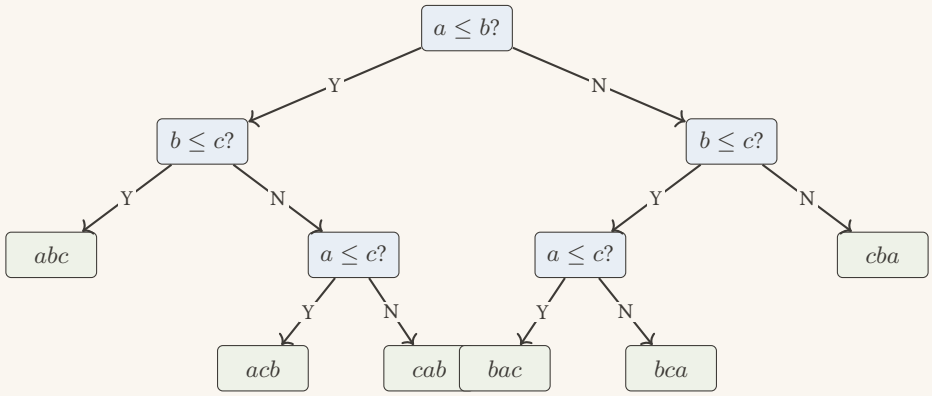
Key idea $\Omega(n \log n)$ comparison-sort lower bound

For any comparison sort to be *correct*, every one of the $n!$ input permutations must reach a distinct leaf — otherwise two inputs that need different answers would be given the same answer. A binary tree with at least $n!$ leaves has height at least $\lceil \log_2(n!) \rceil$. By Stirling's approximation, $\log_2(n!) = n \log_2 n - n \log_2 e + O(\log n) = \Theta(n \log n)$. Therefore any comparison sort makes $\Omega(n \log n)$ comparisons in the worst case.

The proof is short. Stirling's approximation gives $n! \geq (n/e)^n$, so $\log_2(n!) \geq n \log_2 n - n \log_2 e \geq \frac{1}{2} n \log_2 n$ for all $n \geq 4$. The decision-tree height bound transfers directly to a worst-case-comparisons bound, and worst-case comparisons bound the worst-case running time from below.

The 3-element decision tree

The smallest non-trivial example: $n = 3$. There are $3! = 6$ permutations, so the tree must have at least 6 leaves, and a binary tree with 6 leaves has height at least $\lceil \log_2 6 \rceil = 3$. The optimal sort for $n = 3$ uses exactly 3 comparisons in the worst case — and the tree below is essentially what *insertion sort* on 3 elements builds.



The leaves are the 6 permutations of $\{a, b, c\}$. The tree has height 3, matching $\lceil \log_2 6 \rceil$. No 3-element comparison sort can do better in the worst case: any tree with fewer than 6 leaves would alias two distinct permutations and be incorrect.

Comparison vs. non-comparison sorts

Aside Comparison vs. non-comparison

The $\Omega(n \log n)$ bound applies *only* to sorts that use key comparisons as their sole inspection of the input. Sorts that look *inside* keys — counting sort treats the key as an index, radix sort treats it as a sequence of digits, bucket sort treats it as a numeric value to bin — bypass the bound entirely. This is not a contradiction: those sorts have access to extra structure (key universe size, digit count, distribution assumption) that a generic comparison sort does not. The non-comparison sorts of §13.6–§13.7 trade that extra-knowledge requirement for sub-$n \log n$ time.
The decision tree of a comparison sort fans out by 2 at each internal node ($\leq$ vs. $>$). A counting sort on an n-element input over a key universe of size k has access to k-way branching at the indexing step, which is what lets it land permutations in linear time when $k = O(n)$.

2 13.1 Bubble Sort

The algorithm

Definition —

Bubble sort. Repeatedly walk the array, comparing adjacent pairs and swapping if out of order. After each pass, the largest remaining unsorted element "bubbles" to the end.

```
void bubbleSort(std::vector<int>& a) {
    int n = a.size();
    for (int i = 0; i < n - 1; ++i) {
        for (int j = 0; j < n - i - 1; ++j) {
            if (a[j] > a[j + 1])
                std::swap(a[j], a[j + 1]);
        }
    }
}
```

The inner loop shrinks each pass because the last i elements are already in their final position.

Cost

- Best case (already sorted): still $O(n^2)$ comparisons in the plain version. An optimization: short-circuit when an entire pass made no swaps, giving $O(n)$ best case.
- Average: $O(n^2)$.
- Worst case (reverse sorted): $O(n^2)$ comparisons, $O(n^2)$ swaps.
- Space: $O(1)$.
- Stable: yes (equal elements never swap past each other).

Aside Bubble sort vs. the lower bound of §13.0

Bubble sort's $O(n^2)$ average and worst-case cost is a long way above the comparison-sort lower bound of $\Omega(n \log n)$ from §13.0 — bubble is one of the worst-performing comparison sorts that still gets the answer right. The chapter-3 sorts (insertion / selection / merge) sit at the same $O(n^2)$ ceiling (insertion / selection) or at the optimal $O(n \log n)$ bound (merge); ch. 7's heapsort also achieves $O(n \log n)$. So the ladder runs: bubble / insertion / selection at the bottom ($O(n^2)$); merge / heap / introsort / Timsort / pdqsort at the optimal floor ($O(n \log n)$); counting / radix / bucket below the floor for suitable data. The chapter-13 closing decision table in §13.9 collects all of these by problem.

Why you'll never use bubble sort

Gotcha

Bubble sort is a **teaching tool**, nothing more. For any real data: `std::sort` (introsort), `std::stable_sort` (merge sort), or a radix/bucket sort for integers. Bubble sort's $O(n^2)$ worst case on large inputs is not just slow – it's catastrophically slow. $n = 10^6$ is 10^{12} operations, roughly an hour of CPU time. A real sort does the same job in milliseconds.

Aside

The one defense of bubble sort: its best case with the short-circuit optimization is $O(n)$, matching insertion sort. For nearly-sorted data, it's not terrible. But insertion sort is better on the same workload and easier to reason about, so even that niche has a better answer. (Insertion / selection / merge sort live in ch. 3; heapsort in ch. 7. §13.9's decision table collects every option side-by-side.)

What bubble sort is good for

- Teaching the concept of an in-place comparison sort.
- Interview questions about algorithmic complexity.
- Tiny inputs ($n \leq 20$) where the constant factor is irrelevant.
- Hardware sorting networks, where adjacent-pair compare-and-swap is the primitive operation.

3 13.2 Quickselect

The problem

Definition —

k-th smallest element. Given an array and an integer k, return the k-th smallest value. For $k = 0$, the minimum; for $k = n - 1$, the maximum; for $k = n/2$, the median.

The naive solution: sort, return `a[k]`. That's $O(n \log n)$. Quickselect does it in $O(n)$ average time *without* fully sorting.

The insight

Quickselect borrows Quicksort's partition routine – the one that picks a pivot, puts smaller elements to the left, larger to the right, and returns the pivot's final position.

Key idea

Quicksort recurses on both sides; quickselect recurses on only the side that contains the k-th element. That cuts the total work from $O(n \log n)$ to $O(n)$ on average, because the unused half of each partition is discarded.

Algorithm

```
int partition(std::vector<int>& a, int lo, int hi) {
    int pivot = a[(lo + hi) / 2];
    int i = lo - 1, j = hi + 1;
    while (true) {
        do { ++i; } while (a[i] < pivot);
        do { --j; } while (a[j] > pivot);
        if (i >= j) return j;
        std::swap(a[i], a[j]);
    }
}

int quickselect(std::vector<int>& a, int lo, int hi, int k) {
    if (lo >= hi) return a[lo];
    int p = partition(a, lo, hi);
    if (k <= p) return quickselect(a, lo, p, k);
    return quickselect(a, p + 1, hi, k);
}
```

Cost

- **Average**: $O(n)$. Partition costs n; the next recursion is on $n/2$; then $n/4$; …total $n + n/2 + n/4 + \ldots = O(n)$ by geometric sum.
- **Worst**: $O(n^2)$, same as naive quicksort, when pivots are consistently bad.
- **Median-of-medians** gives $O(n)$ worst-case selection, but the constant is large; in practice quickselect with a random or median-of-three pivot is faster.

Median-of-medians for worst-case $O(n)$ selection

The median-of-medians algorithm (Blum, Floyd, Pratt, Rivest, Tarjan, 1973; CLRS §9.3) gives *worst-case* $O(n)$ selection by computing a pivot guaranteed to be "good enough" on every input. The construction:

1. Split the input into $\lceil n/5 \rceil$ groups of 5 consecutive elements (the last group may have fewer than 5).
2. Sort each group in $O(1)$ time (5 elements: at most 7 comparisons by an optimal 5-element sorting network). The middle of each sorted group is that group's median.
3. *Recursively* call median-of-medians on the collected $\lceil n/5 \rceil$ group-medians to find their median — call it M.
4. Use M as the pivot for the standard quickselect partition step.

The pivot M is greater than at least $3\lceil n/10 \rceil - 6$ elements and less than at least $3\lceil n/10 \rceil - 6$ elements

(each of the $\lceil n/10 \rceil$ groups whose median is below M contributes at least 3 elements below M). This guarantees that every recursive call shrinks the problem by at least a $7/10$ factor in the worst case. The recurrence $T(n) \leq T(n/5) + T(7n/10) + O(n)$ solves to $T(n) = O(n)$ because $1/5 + 7/10 = 9/10 < 1$.

Aside `std::nth_element` guarantees

The C++ standard's `std::nth_element` is required to run in average linear time but does not guarantee worst-case linear time. libstdc++ implements it as introselect: quickselect with a recursion-depth fallback to median-of-medians (analogous to `std::sort`'s introsort fallback to heapsort). In practice the recursion-depth fallback is rarely hit on production data — random pivots make the worst case astronomically unlikely — but the fallback exists so the worst case is bounded.

C++ standard library

You don't have to write quickselect yourself – C++ gives it to you:

```
#include <algorithm>
// Rearranges [begin, end) so nth is the element that would be at
// position n in a fully sorted range. Everything before nth is <=,
// everything after is >=. Average O(n).
std::nth_element(a.begin(), a.begin() + k, a.end());
int kthSmallest = a[k];
```

Key idea

`std::nth_element` is the correct answer to "what's the median / top-k / k-th percentile" in any interview or real codebase. It's linear time, in place, and a single line of code. Reach for it before thinking harder.

Example Median of $[3, 1, 4, 1, 5, 9, 2, 6]$ via quickselect

Input: $a = [3, 1, 4, 1, 5, 9, 2, 6]$, $n = 8$, target $k = n/2 = 4$ (the upper median — the 5th-smallest, 0-indexed position 4).
Sorted: $[1, 1, 2, 3, 4, 5, 6, 9]$ so the answer is 4.
Trace using the listing above's Hoare partition with midpoint pivot:
Call 1: `quickselect(a, 0, 7, 4)`. Pivot = $a[(0+7)/2] = a[3] = 1$. Hoare's two-pointer scan: i advances from -1 while $a[i] < 1$ (no, $a[0] = 3 \geq 1$), j retreats from 8 while $a[j] > 1$ ($a[7] = 6, a[6] = 2$, then $a[5] = 9, a[4] = 5, a[3] = 1$, stop). $i = 0, j = 3$. Since $i < j$, swap $a[0] \leftrightarrow a[3]$: $a = [1, 1, 4, 3, 5, 9, 2, 6]$. Continue: i advances to 1 ($a[1] = 1$, not less than pivot, stop), j retreats to 0 ($a[2] = 4, a[1] = 1$, stop). Now $i = 1 \geq j = 1$. Return $j = 1$. Pivot's final position is 1; $k = 4 > 1$, so recurse right: `quickselect(a, 2, 7, 4)`.
Call 2: `quickselect(a, 2, 7, 4)` on subarray $[4, 3, 5, 9, 2, 6]$. Pivot = $a[(2+7)/2] = a[4] = 5$. After partition: subarray becomes $[4, 3, 2, 5, 9, 6]$ (or similar depending on the exact swap order; pivot 5 lands at position 5). Hoare scan returns $j = 5$. $k = 4 \leq 5$, recurse left: `quickselect(a, 2, 5, 4)`.
Call 3: `quickselect(a, 2, 5, 4)` on subarray $[4, 3, 2, 5]$. Pivot = $a[(2+5)/2] = a[3] = 3$. After partition: $[2, 3, 4, 5]$ with pivot at position 3. Returns $j = 3$. $k = 4 > 3$, recurse right: `quickselect(a, 4, 5, 4)`.
Call 4: `quickselect(a, 4, 5, 4)` on $[4, 5]$. Pivot = $a[(4+5)/2] = a[4] = 4$. Partition is no-op (already in order); returns $j = 4$. $k = 4 \leq 4$, recurse left: `quickselect(a, 4, 4, 4)` returns $a[4] = 4$.

Result: the 5th-smallest (0-indexed position 4) is **4**. The total comparison work was $O(n)$ as expected: calls processed subarrays of sizes $8, 6, 4, 2$, summing under the geometric series $\leq 2n$.

Applications

- **Median / percentile computation.**
- **Top-k queries.** After `nth_element`, the first k entries are the top-k (unsorted). If you need them sorted, sort just that prefix.
- **Machine learning**: quickselect-based k-nearest-neighbor scoring.
- **Streaming statistics**: approximate quantiles via repeated quickselect.

4 13.3 Bucket Sort

The idea

Definition —

Bucket sort. Distribute values into K buckets based on a range partition, sort each bucket with another algorithm, concatenate. $O(n)$ average when values are uniformly distributed.

This is a **non-comparison sort** – it exploits numerical structure of the values to beat the $\Omega(n \log n)$ comparison-sort lower bound (§13.0).

Aside Scope of §13.3 vs. §13.6 / §13.7

This section covers *bucket sort proper* (range-partition into K buckets, per-bucket comparison sort, concatenate). Two sibling sorts in the non-comparison family get their own sections later: *counting sort* (bucket sort degenerated to one bucket per key value, no per-bucket sort needed) lives in §13.6, and *radix sort* (iterated counting sort, one pass per digit) lives in §13.7. The closing notebox of this section sketches the family-tree relationship; the full algorithms come later.

Algorithm

Given non-negative numbers up to some max value M:

1. Create K buckets. Bucket i holds values in $[i \cdot M/K, (i+1) \cdot M/K)$.
2. For each element x, compute `bucket = floor(x * K / (M + 1))` and append x to that bucket.
3. Sort each bucket (insertion sort or any other; each bucket is small).
4. Concatenate buckets in order.

```
void bucketSort(std::vector<int>& a, int K) {
    if (a.empty()) return;
    int maxV = *std::max_element(a.begin(), a.end());
    std::vector<std::vector<int>> buckets(K);

    for (int x : a) {
```

```
        int idx = (long long)x * K / (maxV + 1);
        buckets[idx].push_back(x);
    }

    a.clear();
    for (auto& b : buckets) {
        std::sort(b.begin(), b.end());
        a.insert(a.end(), b.begin(), b.end());
    }
}
```

Cost

Let n be the array size and K the bucket count.

- **Distribution pass**: $O(n)$.
- **Per-bucket sort**: if values are uniform, each bucket has $\approx n/K$ elements. Sorting each with $O(m^2)$ insertion sort gives total $K \cdot (n/K)^2 = n^2/K$. With $K = n$, this is $O(n)$.
- **Concatenate**: $O(n + K)$.
- **Total (uniform distribution, $K = n$)**: $O(n)$.
- **Worst case (all elements in one bucket)**: $O(n^2)$.

Gotcha

Bucket sort's $O(n)$ is *contingent on distribution*. If all your values cluster in a narrow range (e.g., 100 random integers between 1 and 10^9), bucket sort degenerates: 99 empty buckets, one bucket with everything. You're paying for K buckets to do nothing. Use bucket sort only when you know the data is roughly uniform over a known range.

Related algorithms

- **Counting sort**: degenerate bucket sort where each bucket holds exactly one value. $O(n + M)$ time, $O(M)$ space. Perfect for small-range integers. Full treatment in §13.6.
- **Radix sort**: iterate counting sort on each digit. $O(d(n + b))$ for d-digit numbers in base b. Used in database sorts, GPU sorts, and networking. Full treatment in §13.7.
- **Pigeonhole sort**: counting sort by another name, when value range = number of elements.

Key idea

The $n \log n$ barrier only applies to comparison sorts. Any sort that uses numeric / string structure of its inputs – bucket, radix, counting – can beat it for suitable data. Production string sorting (timsort in Python, dpqsort in Java) is comparison-based; production integer sorting (std::sort in libstdc++ at large sizes) can fall through to radix for speed.

5 13.4 List Data Structure

The idiom

A linked list (ch. 4) can be represented in two ways:

- **Just a head pointer** (and maybe a tail pointer), held in local variables or as member fields.
- **A wrapper container** – a `List` struct that owns `head`, `tail`, and maybe `size`. Functions take a `List*` instead of a raw head pointer.

```
// Approach 1: just nodes.
struct Node { int data; Node* next; };
Node* head = nullptr;
Node* tail = nullptr;
int size = 0;

// Approach 2: list wrapper.
struct List {
    Node* head = nullptr;
    Node* tail = nullptr;
    int size = 0;
};

void listPushBack(List* lst, int data) {
    Node* n = new Node{data, nullptr};
    if (!lst->head) lst->head = n;
    else            lst->tail->next = n;
    lst->tail = n;
    ++lst->size;
}
```

Why wrap?

Key idea

The wrapper exists so that list-manipulating functions take *one* parameter instead of three (head, tail, size). More importantly, operations like "insert at tail" or "get length" become $O(1)$ because the struct already knows what the naked-node version would have to recompute.

Operation	Head-only list	Wrapped list
Insert at head	$O(1)$	$O(1)$
Insert at tail	$O(n)$ (walk to end)	$O(1)$ (tail ptr)
Get size	$O(n)$ or maintained externally	$O(1)$ (cached)
Pass to function	3 args (head, tail, size)	1 arg (`List*`)

Why not just use `std::list`?

Gotcha

In C++, `std::list` is the wrapper-style doubly-linked list and you should use it unless you have a specific reason not to. Writing your own list is a pedagogical exercise; in production, `std::list` (or `std::forward_list` for singly-linked) is tested, debuggers know how to render it, and you don't leak memory on exception. The "list data structure" the textbook describes is reinventing `std::list`.

Aside `std::list` vs. `std::forward_list`

The C++ STL ships *two* linked-list containers, mirroring the two flavours covered in ch. 4:

- `std::list` (since C++98) is a *doubly-linked list* with head and tail sentinels: $O(1)$ `push_front` / `pop_front` / `push_back` / `pop_back`, $O(1)$ `size()` (cached since C++11), $O(1)$ insertion and deletion at any iterator. Iterator invariants are stable: an iterator to a node is valid for that node's entire lifetime, not just until the next mutation.
- `std::forward_list` (since C++11) is a *singly-linked list* with a head-only sentinel: $O(1)$ `push_front` / `pop_front`, $O(n)$ tail operations (no tail pointer, by design), *no* `size()` method (would require $O(n)$ traversal — C++11 left it off rather than cache it, since the use case for `forward_list` is "smaller per-node footprint when you don't need cached size or doubly-linked traversal").

The naming follows the same pattern as `std::set` / `std::unordered_set` (ch. 12): the unqualified name is the more featureful default.

When linked lists are the wrong answer

Even with a nice wrapper, linked lists are frequently worse than `std::vector` (dynamic array):

- Cache locality: array elements are contiguous, list nodes scattered across the heap.
- Memory overhead: each node has a `next` pointer, often doubling memory vs. raw values.
- Random access: $O(n)$ for list, $O(1)$ for array.
- Iteration: much slower on a list due to cache misses on every `next`.

Use a linked list when you need cheap insertion/deletion at interior points with existing iterators, or when you're building something that requires pointer stability (e.g., an LRU cache with external references). Otherwise, default to `std::vector`.

6 13.5 Circular Lists

The definition

Definition —

Circular linked list. A linked list (ch. 4) where the tail's `next` pointer points back to the head, not `nullptr`. Traversal wraps forever unless explicitly bounded.

```
// Singly-linked circular
struct Node { int data; Node* next; };
// Last node's next points back to head.
```

```
// Doubly-linked circular (most common in practice)
struct DNode { int data; DNode* prev; DNode* next; };
// Last node's next -> head; head's prev -> last node.
```

Traversal

The key trick: stop when you come back to the start, not when you hit `nullptr`.

```
void traverseCircular(Node* head) {
    if (!head) return;
    Node* cur = head;
    do {
        visit(cur);
        cur = cur->next;
    } while (cur != head);
}
```

Gotcha

Use do-while, not while. A regular `while (cur != head)` skips the head on the first iteration because we start at head and the condition fails immediately. The do-while runs the body first, then checks the condition.

Why circular lists exist

- **Round-robin scheduling.** OS schedulers cycle through ready processes; each yield pops one and reappends it at the end. A circular list gives O(1) "next process" and O(1) append.
- **Music players on "repeat all."** Queue is a circular list of tracks.
- **Ring buffers** (logically circular, though usually array-based for cache locality).
- **Josephus problem** and other classic cyclic-elimination algorithms – naturally circular.
- **Token-passing networks.** Token Ring Ethernet was literally a circular list of nodes; each station had one successor.
- **Doubly-linked circular lists with a sentinel** give branch-free insert/delete at any position.

Sentinel trick for doubly-linked circular lists

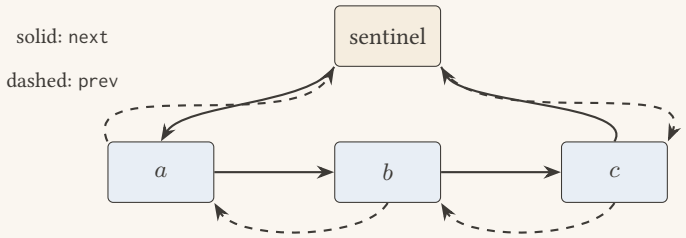
A *sentinel* node – a dummy that's never "real" data but completes the circle – removes the "is this node the head / tail" special cases:

```
struct DNode { int data; DNode* prev; DNode* next; };

struct CircularList {
    DNode sentinel;   // sentinel.next = head, sentinel.prev = tail
    CircularList() { sentinel.prev = sentinel.next = &sentinel; }
};
```

```
// Insert n before 'where' -- no null checks needed.
void insertBefore(DNode* where, DNode* n) {
    n->next = where;
    n->prev = where->prev;
    where->prev->next = n;
    where->prev = n;
}

// Remove n -- no null checks needed.
void remove(DNode* n) {
    n->prev->next = n->next;
    n->next->prev = n->prev;
}
```



The sentinel is always present, even when the list is logically empty (then `sentinel.next == sentinel.prev == &sentinel`). There is no "`head == nullptr`" branch in `insertBefore` / `remove` — the sentinel guarantees both `prev` and `next` are non-null for every real node, and operations on the sentinel itself are well-defined.

Key idea

The sentinel technique eliminates all "empty list" and "at the head" edge cases. Insert/remove are six pointer writes with no branching. The Linux kernel's `list_head` uses exactly this structure, and the design has been copied into countless production codebases – it's one of the most elegant tricks in systems programming.

Traversal in reverse (doubly-linked)

Mirror of the forward case – walk `prev` pointers from any starting point until you return:

```
void traverseReverseCircular(DNode* tail) {
    if (!tail) return;
    DNode* cur = tail;
    do {
        visit(cur);
        cur = cur->prev;
    } while (cur != tail);
}
```

7 13.6 Counting sort

The idea

Definition —

Counting sort. A linear-time stable sort for keys in $[0, k)$. Counts the occurrences of each key value, computes a prefix-sum to find each key's starting output position, then places each input element into the output array in a single right-to-left pass. Runs in $O(n + k)$ time and $O(n + k)$ space.

Counting sort sidesteps the comparison-sort lower bound of §13.0 by treating each key as an array index rather than as something to compare. The price: it only works when keys are non-negative integers in a known small-ish range.

Algorithm

Given input $A[0 \ldots n-1]$ with each $A[i] \in [0, k)$:

1. **Count.** Allocate $C[0 \ldots k-1]$ initialised to zero. Scan A once; for each $A[i]$, increment $C[A[i]]$. Now $C[v]$ is the number of input elements equal to v.
2. **Prefix sum.** Replace C with its running sum: $C[v] \leftarrow C[v] + C[v-1]$ for v from 1 to $k-1$. Now $C[v]$ is the number of input elements $\leq v$, which is also one past the last position where a v should land in the output.
3. **Place.** Allocate output $B[0 \ldots n-1]$. Iterate i from $n-1$ down to 0 (*right-to-left* for stability); for each $A[i]$, set $C[A[i]] \leftarrow C[A[i]] - 1$, then $B[C[A[i]]] \leftarrow A[i]$.

```
// Stable counting sort. Keys must satisfy 0 <= a[i] < k.
void countingSort(std::vector<int>& a, int k) {
    int n = a.size();
    if (n == 0) return;
    std::vector<int> C(k, 0);
    std::vector<int> B(n);

    // 1. Count.
    for (int i = 0; i < n; ++i) ++C[a[i]];

    // 2. Prefix sum: C[v] = number of inputs <= v.
    for (int v = 1; v < k; ++v) C[v] += C[v - 1];

    // 3. Place right-to-left for stability.
    for (int i = n - 1; i >= 0; --i) {
        --C[a[i]];
        B[C[a[i]]] = a[i];
    }

    a = std::move(B);
}
```

Cost

- **Time**: $O(n + k)$. Three linear scans of size n, k, n.

- **Space**: $O(n + k)$. The count array C has size k; the output array B has size n.
- **Stable**: yes. The right-to-left placement guarantees that two equal elements preserve their input order. (Going left-to-right would reverse them — this is the most common counting-sort bug.)
- **In-place**: no. Counting sort needs $O(n + k)$ auxiliary space.

Key idea When counting sort beats quicksort

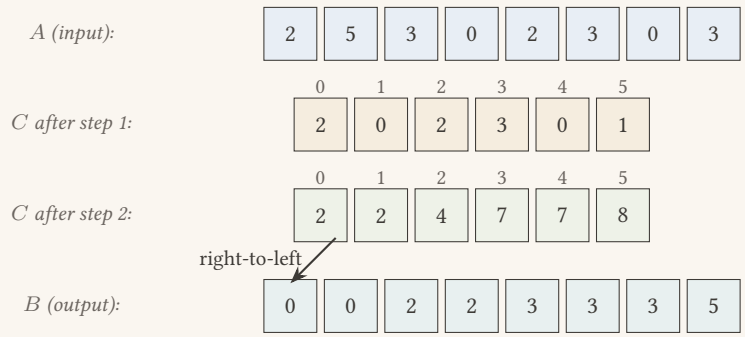
Counting sort outperforms quicksort exactly when $k = O(n)$ — i.e. the key universe is at most linear in the input size. At $k = n$, counting sort is $O(n)$ vs. quicksort's $O(n \log n)$. At $k = n^2$, counting sort costs $O(n^2)$ for the prefix-sum scan alone — worse than *any* comparison sort. The break-even crossover is roughly $k = n \log n$. As a rule of thumb: counting sort beats comparison sorts when the key universe is at most a few times the input size and you can afford the $O(k)$ memory for the count array.

Stability proof

Claim. Counting sort with right-to-left placement is stable: for $i < j$ with $A[i] = A[j]$, the output places $A[i]$ before $A[j]$.

Proof sketch. The right-to-left placement loop processes $A[j]$ first (since $j > i$), decrements $C[A[j]]$, and writes $A[j]$ to position $C[A[j]]$. When the loop later reaches $A[i]$, $C[A[i]]$ has been decremented at least once (by the $A[j]$ write), so $A[i]$ writes to a smaller index — i.e. to the left of $A[j]$ in the output. Stability holds for any equal-key pair.

Worked layout: counting array → output



The picture above sorts $A = [2, 5, 3, 0, 2, 3, 0, 3]$ in three phases: the count vector C tallies each key value (top: two 0s, no 1s, two 2s, three 3s, no 4s, one 5); the prefix sum ($C \leftarrow$ running sum) gives the position one past where each key class ends in B; the right-to-left placement step walks the input backward and stamps each element into its final slot, decrementing C as it goes.

Example Counting sort on $[2, 5, 3, 0, 2, 3, 0, 3]$

$A = [2, 5, 3, 0, 2, 3, 0, 3]$, $n = 8$, $k = 6$ (keys in $[0, 6)$).
Step 1 (count). $C = [\,2,\ 0,\ 2,\ 3,\ 0,\ 1\,]$ — two 0s, no 1s, two 2s, three 3s, no 4s, one 5.
Step 2 (prefix sum). $C = [\,2,\ 2,\ 4,\ 7,\ 7,\ 8\,]$. Each $C[v]$ is now "one past the last position where a v goes." For example, the 3s end at output position 7, so $B[4..6]$ will hold the three 3s.
Step 3 (right-to-left placement).

- $i = 7$: $A[7] = 3$. $C[3] \to 6$, $B[6] = 3$.
- $i = 6$: $A[6] = 0$. $C[0] \to 1$, $B[1] = 0$.
- $i = 5$: $A[5] = 3$. $C[3] \to 5$, $B[5] = 3$.
- $i = 4$: $A[4] = 2$. $C[2] \to 3$, $B[3] = 2$.
- $i = 3$: $A[3] = 0$. $C[0] \to 0$, $B[0] = 0$.
- $i = 2$: $A[2] = 3$. $C[3] \to 4$, $B[4] = 3$.
- $i = 1$: $A[1] = 5$. $C[5] \to 7$, $B[7] = 5$.
- $i = 0$: $A[0] = 2$. $C[2] \to 2$, $B[2] = 2$.

Result. $B = [0, 0, 2, 2, 3, 3, 3, 5]$, sorted and stable. Total work: $n = 8$ for the count pass, $k = 6$ for the prefix-sum pass, $n = 8$ for placement.

Use cases

- **Pre-pass for radix sort.** Each digit pass of radix sort is a counting sort on $k = b$ (the radix). §13.7 fully exploits this.
- **Integer histograms in image processing.** Pixel intensities are 8-bit ($k = 256$); a counting-sort "count" phase *is* the histogram, and the prefix-sum phase *is* the cumulative distribution function used in histogram equalisation.
- **Database aggregations.** `GROUP BY` on a low-cardinality column (e.g. enum / boolean / status) is a counting sort: bucketise rows by group key in one pass.
- **Bucket sort sub-step.** When per-bucket sort in §13.3 turns out to be on a small key universe, counting sort is what to reach for.

8 13.7 Radix sort

The idea

Definition —

Radix sort. Sort d-digit keys in base b by repeating a stable single-digit sort once per digit. The standard form is *LSD radix sort*: process digits from least-significant to most-significant, with counting sort doing each digit pass. Runs in $O(d(n + b))$ time and $O(n + b)$ space.

Radix sort beats the comparison-sort lower bound when keys are fixed-width: $O(d(n + b)) = O(n)$ if d and b are both constants. For 32-bit integers with $b = 256$ ($d = 4$ byte passes), that gives roughly $4 \cdot (n + 256) = O(n)$ work — which is why radix sort is the standard parallel sort on GPUs and the go-to integer sort in production analytics engines.

LSD algorithm

The crucial property: each digit pass must be a *stable* sort, and counting sort is stable by construction (§13.6). So:

1. For digit position $p = 0, 1, \ldots, d - 1$ (from least-significant to most-significant):
   (a) Stable-counting-sort the array on the value of digit p (i.e. $\lfloor x/b^p \rfloor \bmod b$ as the key for input x).

2. After d passes, the array is fully sorted.

```
// LSD radix sort over non-negative integers in base b.
// d is the number of base-b digits sufficient to represent
// every input element.
void radixSortLSD(std::vector<int>& a, int b, int d) {
    int n = a.size();
    if (n == 0) return;
    std::vector<int> B(n);
    long long base = 1;  // = b^p

    for (int p = 0; p < d; ++p) {
        std::vector<int> C(b, 0);

        // Count digit-p occurrences.
        for (int i = 0; i < n; ++i) {
            int digit = (a[i] / base) % b;
            ++C[digit];
        }
        // Prefix sum.
        for (int v = 1; v < b; ++v) C[v] += C[v - 1];
        // Stable placement, right-to-left.
        for (int i = n - 1; i >= 0; --i) {
            int digit = (a[i] / base) % b;
            --C[digit];
            B[C[digit]] = a[i];
        }
        std::swap(a, B);
        base *= b;
    }
}
```

Cost

- **Time**: $O(d(n+b))$. Each of the d digit passes is a counting sort with parameters n and b.
- **Space**: $O(n+b)$. One output buffer of size n plus one count buffer of size b, reused across passes.
- **Stable**: yes. Stability of each digit pass is required for correctness.
- **In-place**: no, in the LSD form. (MSD variants can be made closer to in-place at the cost of recursion overhead.)

Base-vs-passes tradeoff

Key idea Choosing the radix base b

The cost $O(d(n+b))$ is minimised by trading d against b. For w-bit keys: $d = w/\log_2 b$ passes. Doubling b halves the number of passes but doubles the count-array size. The practical sweet spot for 32-bit / 64-bit integer keys is $b = 256$ — one byte per pass, fits comfortably in L1 cache, and the count array is $256 \cdot 4 = 1024$ bytes (or $256 \cdot 8 = 2048$ bytes for 64-bit counters), tiny relative to any production input. Production radix sorts (CUB on GPU, Boost `spreadsort`, `ips4o`'s integer fast-path)

all use byte-per-pass radix.

LSD passes on $[170, 45, 75, 90, 802, 24, 2, 66]$

Input:	170	45	75	90	802	24	2	66
After ones:	170	90	802	2	24	45	75	66
After tens:	802	2	24	45	66	170	75	90
After hundreds:	2	24	45	66	75	90	170	802

Example LSD radix sort on $[170, 45, 75, 90, 802, 24, 2, 66]$

Input: $[170, 45, 75, 90, 802, 24, 2, 66]$. Base $b = 10$, three digits sufficient ($d = 3$, since the max value 802 has 3 digits).

Pass 1 — ones digit. The ones digits are $0, 5, 5, 0, 2, 4, 2, 6$. Stable-counting-sort by ones digit:

$$[170, \mathbf{90}, 802, 2, 24, 45, 75, 66]$$

(170 keeps its position relative to 90 even though both have ones-digit 0, because 170 came before 90 in the input. Then 802 and 2 with ones-digit 2 in input order; 24 alone with 4; 45 before 75 in input order with 5; 66 alone with 6.)

Pass 2 — tens digit. Tens digits of the pass-1 output are $7, 9, 0, 0, 2, 4, 7, 6$. Stable-counting-sort by tens digit:

$$[802, 2, 24, 45, 66, 170, 75, 90]$$

(802 and 2 have tens-digit 0 and stay in pass-1 order; 24, 45, 66 have tens 2, 4, 6; 170 and 75 both have tens-digit 7 and stay in pass-1 order; 90 has tens-digit 9.)

Pass 3 — hundreds digit. Hundreds digits of the pass-2 output are $8, 0, 0, 0, 0, 1, 0, 0$. Stable-counting-sort by hundreds digit:

$$[2, 24, 45, 66, 75, 90, 170, 802]$$

(All seven hundreds-digit-0 values stay in their pass-2 order; 170 has hundreds-digit 1; 802 has hundreds-digit 8.)

Result. Fully sorted in $d = 3$ passes, each pass an $O(n + b)$ counting sort. Total work for $n = 8$, $b = 10$: $3 \cdot (8 + 10) = 54$ key inspections plus the placement loops — linear in n.

MSD radix sort and variable-length keys

LSD radix sort requires every key to have the same number of digits (or be left-padded with zeros). For variable-length keys (strings of different lengths, IP addresses with optional prefixes, etc.) the natural variant is *MSD radix sort*: process digits from *most*-significant to least-significant, recursing into each non-empty bucket.

- **Algorithm.** Bucket the input by the most-significant digit. Recursively radix-sort each non-empty bucket on the next digit. Concatenate buckets in MSD order.
- **Strings.** MSD radix on strings is the standard string-sort: bucket by first character, recurse on remaining suffix. The recursion bottoms out when a bucket has ≤ 1 string (already sorted) or when the suffix is empty.
- **When MSD beats LSD.** Variable-length keys (LSD would need padding); early-termination when most strings differ in their first few characters; cache locality (each MSD bucket fits in cache).
- **When LSD beats MSD.** Fixed-width keys (no recursion overhead); simpler code; no stack-depth bound.

Use cases

- **GPU sorts.** Radix sort is the canonical parallel sort on GPUs (CUB, Thrust, NVIDIA's `cub::DeviceRadixSort`). The pattern is fully data-parallel: each digit pass is a histogram + prefix-scan + scatter, all with well-known GPU primitives.
- **Database engine sorts.** PostgreSQL's `ORDER BY` on integer / fixed-width keys; BigQuery / Snowflake / DuckDB column-store sorts.
- **Spotify / streaming-recommendation pipelines.** Sort-by-user-id or sort-by-track-id at the petabyte scale uses radix because integer key universes are natural and the parallelism is trivial.
- **Network packet processing.** Sort by source IP / destination IP / 5-tuple flow key is a fixed-width radix sort, used in DPDK and packet-capture pipelines.

9 13.8 Production hybrid sorts

The lower bound of §13.0 says no comparison sort beats $O(n \log n)$ in the worst case. But real data is not the worst case — it has structure (already-sorted runs, nearly-sorted prefixes, repeated keys, small slices), and the performance constants vary by an order of magnitude across algorithms (cache behaviour, branch prediction, allocator behaviour). Every modern standard library ships a *hybrid sort*: a primary algorithm (quicksort variant), a fallback for adversarial inputs, and a small-array fast-path. This section covers the four most-deployed hybrids.

Introsort

Definition —

Introsort (David Musser, 1997). Quicksort with two fallbacks: (a) when recursion depth exceeds $2 \log_2 n$, switch to heapsort on the current partition; (b) when partition size drops below a threshold (typically 16–32 elements), switch to insertion sort. Worst case $O(n \log n)$, typical case faster than vanilla quicksort.

- **Why the heapsort fallback.** Vanilla quicksort degrades to $O(n^2)$ on adversarial inputs (already sorted, all equal keys with a naive pivot). Heapsort (ch. 7) is $O(n \log n)$ worst case and in-place; the $2 \log_2 n$ recursion-depth threshold ensures the fallback triggers before quicksort does too much damage.
- **Why the insertion-sort fast-path.** Insertion sort is $O(n^2)$ asymptotically but has a small constant and is cache-friendly; for small ranges ($n \leq 16$ or so) it is empirically faster than quicksort due to lower

per-call overhead.

- **Where it lives.** `std::sort` in libstdc++ (GCC), libc++ (LLVM), and MSVC — the industry-default C++ general-purpose sort.

Timsort

Definition —

Timsort (Tim Peters, 2002). Hybrid of merge sort and insertion sort that exploits *natural runs* — already-sorted (ascending or strictly descending) subsequences that occur in real data. Detects runs in a single pass, then merges them with a galloping merge. $O(n)$ on already-sorted input, $O(n \log n)$ general, stable.

- **Run detection.** A single forward scan identifies maximal monotone runs (descending runs are reversed in place to make them ascending). Short runs are extended by insertion sort to a minimum length (typically 32–64) so the merge phase has bounded run count.
- **Run-stack merging.** Runs are merged in a stack-based discipline that maintains an invariant on the run-length sequence (each run is either at least twice the next or at most equal — this caps the merge depth at $O(\log n)$).
- **Galloping merge.** When one run dominates the other (one source feeds many consecutive output elements), the merge switches from element-wise compare to exponential-search, dropping per-element cost from $O(\log n)$ to $O(\log k)$ where k is the run-of-wins length.
- **Where it lives.** Python `list.sort` / `sorted` (since 2.3, 2002); Java `Arrays.sort` for objects (since Java 7, 2011); Java `Collections.sort`; Rust's stable `sort` (Timsort variant); Android, V8.

Dual-pivot quicksort (dpqsort)

Definition —

Dual-pivot quicksort (Vladimir Yaroslavskiy, 2009). A quicksort variant that picks *two* pivots per partition, splitting the array into three regions ($< p_1$, $p_1 \leq \cdot \leq p_2$, $> p_2$) and recursing on all three. Empirically about 5% faster than single-pivot quicksort on primitive-integer inputs due to better branch prediction and cache behaviour.

- **Pivot selection.** Sample 5 elements from the partition (positions $\lfloor n/7 \rfloor$, $\lfloor 2n/7 \rfloor$, $\lfloor n/2 \rfloor$, $\lfloor 5n/7 \rfloor$, $\lfloor 6n/7 \rfloor$); use the 2nd and 4th in sorted order as p_1, p_2.
- **Three-way partition.** Single pass with three pointers; the equal-to-pivot middle region needs no recursion when the pivots happen to equal.
- **Where it lives.** Java `Arrays.sort` for primitive types (`int[]`, `long[]`, etc., since Java 7, 2011). Java's two-track sort policy (Timsort for objects, dpqsort for primitives) reflects the different optimisation surface: object sorts dominated by comparison cost, primitive sorts dominated by cache and branching.

Pdqsort

Definition —

Pdqsort (Pattern-Defeating Quicksort, Orson Peters, 2016). An introsort descendant with: (a) Block-Quicksort partitioning (branch-free partition using small lookup tables); (b) pattern detection that switches to merge sort when already-sorted patterns are detected; (c) heapsort fallback on recursion depth (same as introsort).

- **BlockQuicksort partition.** Partition with a small lookup buffer to eliminate branches on the element-pivot comparison. Roughly $2\times$ faster than branched partition on random data.
- **Pattern defeat.** Adversarial inputs that cause vanilla quicksort to degrade are detected within $O(\log n)$ recursion levels, and the algorithm switches to merge sort (still $O(n \log n)$ but immune to the pathological case).
- **Where it lives.** Rust's `sort_unstable` (since 1.20, 2017); Boost.Sort `pdqsort`; `cppsort` library; some libc++ implementations of `std::sort`.

Language → default-sort decision

Language / API	Default sort	Underlying algorithm	Stable?
C++ `std::sort`	introsort	qsort + heap + insertion	no
C++ `std::stable_sort`	merge sort	merge	yes
Python `list.sort`	Timsort	merge + insertion + run scan	yes
Java `Arrays.sort(Object[])`	Timsort	merge + insertion + run scan	yes
Java `Arrays.sort(int[])`	dpqsort	dual-pivot quicksort	no
Rust `sort_unstable`	pdqsort	qsort + block-partition + heap	no
Rust `sort`	Timsort variant	merge + insertion + run scan	yes
Go `sort.Sort`	introsort	qsort + heap + insertion	no
Go `sort.Stable`	merge sort	merge	yes

Aside Why every standard library ships a hybrid

None of bubble, selection, plain quicksort, plain merge, or plain heapsort is the right answer on real data. Vanilla quicksort degrades to $O(n^2)$ on adversarial inputs; vanilla merge sort needs $O(n)$ scratch; heapsort thrashes the cache; insertion sort is $O(n^2)$ on random data. The hybrid combinations above are *strictly better* than any of their components: introsort is heap-fallback safe but quicksort-fast on average; Timsort is merge-stable but linear on already-sorted data. The lesson generalises: whenever a tree of competing single-strategy algorithms exists, the production answer is to dispatch between them based on the input's measured properties — the same lesson that makes introselect (§13.2's median-of-medians fallback) and production memcpy (small / medium / large dispatch) win.

10 13.9 Sort algorithm decision table

The master table for picking a sort given problem properties. Each row is a problem characteristic; each cell is the recommended sort with complexity and a chapter pointer.

Problem profile	Recommended sort	Cost
Tiny array, $n \leq 20$	insertion sort (ch. 3)	$O(n^2)$ acceptable
Random data, in-place, fast	introsort (`std::sort`)	$O(n \log n)$
Random data, stable required	Timsort or merge sort	$O(n \log n)$
Nearly-sorted input	Timsort or insertion sort (ch. 3)	$O(n)$ best case
Bounded integer keys, $k = O(n)$	counting sort (§13.6)	$O(n + k)$
Fixed-width integer keys, large	radix sort LSD (§13.7)	$O(d(n + b))$
Variable-length string keys	MSD radix sort (§13.7)	$O(d(n + b))$
Uniform-real distribution, $[0, 1)$	bucket sort (§13.3)	$O(n)$ expected
Find k-th element only	quickselect (§13.2)	$O(n)$ avg
Stable sort + adversarial input ok	merge sort (ch. 3)	$O(n \log n)$
Worst-case bound on sort	heapsort (ch. 7) or introsort	$O(n \log n)$
No extra memory allowed	introsort or heapsort (ch. 7)	$O(n \log n)$
GPU / massively parallel	radix sort LSD on GPU	$O(d \cdot n/p)$
Data does not fit in RAM	external merge sort (post-cs-300)	$O(n \log n)$ I/O
Java primitive array	dpqsort (§13.8)	$O(n \log n)$
Rust `[T]::sort_unstable`	pdqsort (§13.8)	$O(n \log n)$

Aside Reading the decision table

Three questions to ask, in order, when picking a sort:

1. **Is the key universe small or structured?** If yes (bounded integers, fixed-width, uniform reals), counting / radix / bucket beats the comparison-sort floor and is the right answer regardless of stability / memory considerations.
2. **Do I need stability?** If yes, the hybrid choice narrows to Timsort / merge sort. If not, introsort / pdqsort are typically faster.
3. **Do I have an unusual constraint?** Fixed memory ⇒ heapsort or introsort (no $O(n)$ scratch). Massive parallelism ⇒ radix sort on GPU. Out-of-RAM ⇒ external merge sort.

11 13.10 Algorithmic list idioms

ch. 4 introduced linked lists as a structure; §13.4 and §13.5 covered the wrapping idiom and circular variant. This section covers the canonical *algorithms* that operate on linked lists — the cycle-detection, reversal, merge, and pointer-tricks patterns that underlie a large fraction of production list code and interview questions. All five live at $O(n)$ time and $O(1)$ extra space; the idioms are about *how* to achieve those bounds.

Floyd's tortoise-and-hare cycle detection

Definition —

Floyd's tortoise-and-hare. Detect whether a singly-linked list contains a cycle in $O(n)$ time and $O(1)$ space using two pointers: a slow pointer that advances 1 step per iteration, and a fast pointer that advances 2 steps. If the fast pointer reaches `nullptr`, no cycle. If the two pointers ever meet at the same node, a cycle exists — and the meeting point can be used to find both the cycle's length and its start.

Why it works. If a cycle exists with μ pre-cycle nodes (the tail) and λ cycle-length, then once both pointers

are inside the cycle (after at most μ steps), the fast pointer gains 1 modulo λ per step on the slow pointer, and they meet within λ steps.

```
struct Node { int data; Node* next; };

bool hasCycle(Node* head) {
    Node* slow = head;
    Node* fast = head;
    while (fast && fast->next) {
        slow = slow->next;
        fast = fast->next->next;
        if (slow == fast) return true;
    }
    return false;
}
```

Cycle length. Once `slow == fast`, fix `slow` and advance `fast` one step per iteration until they meet again; the number of steps is λ.

```
int cycleLength(Node* meeting) {
    int len = 1;
    Node* p = meeting->next;
    while (p != meeting) { p = p->next; ++len; }
    return len;
}
```

Cycle start. After the meeting, reset one pointer to `head` and advance both at speed 1; they meet at the cycle entry. (Why: at the meeting, `slow` has travelled $\mu + s$ for some $s < \lambda$ inside the cycle; `fast` has travelled $2(\mu + s)$. Their offset modulo λ is $\mu \bmod \lambda$, so a pointer starting at `head` and advancing μ steps lands at the cycle entry — exactly where `slow` also lands when the head-pointer catches up.)

```
Node* cycleStart(Node* head, Node* meeting) {
    Node* p = head;
    Node* q = meeting;
    while (p != q) { p = p->next; q = q->next; }
    return p;
}
```

Example Floyd's on a 7-node list with cycle starting at node 4

List: nodes $1 \to 2 \to 3 \to 4 \to 5 \to 6 \to 7 \to 4$ (i.e. node 7's `next` loops back to node 4). $\mu = 3$ pre-cycle nodes (1, 2, 3), $\lambda = 4$ cycle nodes (4, 5, 6, 7).

Detection trace.

- Step 0: `slow = 1`, `fast = 1`.
- Step 1: `slow = 2`, `fast = 3`.
- Step 2: `slow = 3`, `fast = 5`.
- Step 3: `slow = 4`, `fast = 7`.
- Step 4: `slow = 5`, `fast = 5` (fast goes $7 \to 4 \to 5$). **Met!**

Cycle length. Fix `slow = 5`, advance p from $5 \to 6 \to 7 \to 4 \to 5$. Took 4 steps $\Rightarrow \lambda = 4$. Correct.
Cycle start. Reset p to head (node 1); both pointers advance at speed 1.

- Step 0: `p = 1`, `q = 5`.
- Step 1: `p = 2`, `q = 6`.
- Step 2: `p = 3`, `q = 7`.
- Step 3: `p = 4`, `q = 4`. **Met!**

The cycle starts at node 4. Correct ($\mu = 3$ steps from head to entry).
Total work. Detection: 4 steps. Cycle length: 4 steps. Cycle start: 3 steps. All bounded by $\mu + \lambda$.

In-place list reversal

Reverse a singly-linked list in-place in $O(n)$ time and $O(1)$ extra space. The key trick: maintain three pointers (`prev`, `cur`, `next`) and rewire each node in turn.

```
// Iterative reversal -- O(n) time, O(1) space.
Node* reverse(Node* head) {
    Node* prev = nullptr;
    Node* cur  = head;
    while (cur) {
        Node* next = cur->next;  // save before clobbering
        cur->next  = prev;
        prev       = cur;
        cur        = next;
    }
    return prev;  // new head
}

// Recursive reversal -- O(n) time, O(n) stack.
Node* reverseRecursive(Node* head) {
    if (!head || !head->next) return head;
    Node* newHead = reverseRecursive(head->next);
    head->next->next = head;
    head->next = nullptr;
    return newHead;
}
```

The iterative form is preferred in production: same big-O, no stack-overflow risk on long lists. The recursive form is elegant for interviews.

Sorted merge of two sorted lists

The merge step from merge sort (ch. 3), but on linked-list representation. $O(n + m)$ time, $O(1)$ extra space (no allocations — splice nodes in-place).

```
// Merge two sorted lists into one sorted list.
// Uses a dummy head to simplify the empty-output case.
Node* mergeSorted(Node* a, Node* b) {
    Node dummy{0, nullptr};
    Node* tail = &dummy;
    while (a && b) {
```

```
        if (a->data <= b->data) {
            tail->next = a; a = a->next;
        } else {
            tail->next = b; b = b->next;
        }
        tail = tail->next;
    }
    tail->next = a ? a : b;
    return dummy.next;
}
```

Nth-from-end (two-pointer)

Find the Nth-from-last node of a singly-linked list in a single pass, $O(n)$ time, $O(1)$ space. Trick: advance a *lead* pointer N steps ahead, then advance both pointers in lockstep until lead hits `nullptr`; the trailing pointer is at the Nth-from-last.

```
// Returns the Nth-from-last node, or nullptr if list has < N nodes.
Node* nthFromEnd(Node* head, int N) {
    Node* lead = head;
    for (int i = 0; i < N; ++i) {
        if (!lead) return nullptr;  // list shorter than N
        lead = lead->next;
    }
    Node* trail = head;
    while (lead) {
        lead  = lead->next;
        trail = trail->next;
    }
    return trail;
}
```

The same pattern — two pointers separated by a fixed gap — underlies Floyd's cycle detection (gap $= 1$ but with different speeds), the sliding-window string algorithms, and the "mark-then-find" linked-list patterns generally.

Detecting intersection of two singly-linked lists

Two non-circular singly-linked lists may share a common suffix (a Y-shape). Detect the intersection node in $O(n + m)$ time and $O(1)$ space.

Algorithm 1: length-difference. Walk both lists to measure their lengths. Advance the longer one by $|\text{len}(A) - \text{len}(B)|$ steps. Then advance both in lockstep; the first node where the two pointers coincide is the intersection (or both reach `nullptr`, meaning no intersection).

```
int listLength(Node* head) {
    int n = 0;
    while (head) { ++n; head = head->next; }
    return n;
}

Node* intersect(Node* a, Node* b) {
```

```
    int la = listLength(a), lb = listLength(b);
    while (la > lb) { a = a->next; --la; }
    while (lb > la) { b = b->next; --lb; }
    while (a && b && a != b) { a = a->next; b = b->next; }
    return (a == b) ? a : nullptr;  // a==b==nullptr => no intersection
}
```

Algorithm 2: simultaneous-advance with switch. A slicker variant: advance both pointers; when one hits `nullptr`, restart it at the other list's head. After at most $\text{len}(A) + \text{len}(B)$ steps, the two pointers either meet at the intersection or both reach `nullptr`. The trick aligns the two pointers at distance $|\text{len}(A) - \text{len}(B)|$ implicitly via the swap.

12 13.11 Production references and further reading

Aside Where production code lives

Comparison sorts in production.

- `std::sort` (C++ STL, all major libs) ⇒ introsort (§13.8). Quicksort + heapsort fallback + insertion-sort fast-path.
- `std::stable_sort` (C++ STL) ⇒ merge sort (Timsort variant in some libs).
- `std::list::sort` ⇒ bottom-up merge sort (works on the linked-list structure directly, $O(\log n)$ extra space for the merge stack).
- `std::nth_element` (C++ STL) ⇒ introselect: quickselect with median-of-medians fallback (§13.2).
- Python `list.sort` / `sorted` ⇒ Timsort (§13.8). Tim Peters, 2002.
- Java `Arrays.sort(int[])` / primitive arrays ⇒ dual-pivot quicksort (§13.8). Yaroslavskiy, 2009.
- Java `Arrays.sort(Object[])` / `Collections.sort` ⇒ Timsort.
- Rust `[T]::sort_unstable` ⇒ pdqsort (§13.8). Orson Peters, 2016.
- Rust `[T]::sort` ⇒ Timsort variant.
- Go `sort.Sort` / `sort.Stable` ⇒ introsort / merge sort.

Non-comparison sorts in production.

- Boost `boost::sort::spreadsort` ⇒ radix-based hybrid (counting sort prefix + comparison sort suffix), the Boost go-to integer / float / string sort.
- Boost `boost::sort::flat_stable_sort` ⇒ in-place stable merge variant.
- `ips4o` (in-place super-scalar samplesort, Axtmann et al., 2017) ⇒ research-grade parallel sort, the fastest general-purpose CPU sort on multi-core machines as of writing.
- NVIDIA CUB `cub::DeviceRadixSort` ⇒ GPU radix sort (§13.7). The standard CUDA sort.
- PostgreSQL `ORDER BY` on integer columns ⇒ falls through to radix sort at large input sizes.

Textbook references.

- CLRS Ch 7 — Quicksort (Hoare vs Lomuto partition, randomized analysis, expected vs worst-case bounds).
- CLRS Ch 8 — Sorting in Linear Time (§8.1 decision-tree lower bound; §8.2 counting sort; §8.3 radix sort; §8.4 bucket sort).
- CLRS Ch 9 — Medians and Order Statistics (§9.2 quickselect; §9.3 median-of-medians).

- Knuth, *The Art of Computer Programming, Volume 3: Sorting and Searching* (2nd ed, 1998). The authoritative deep dive on every sort named in this chapter, plus dozens this chapter does not cover.
- Sedgewick, *Algorithms* (4th ed, 2011), Ch 2. Companion treatment with extensive empirical benchmarking.

Key idea Chapter takeaway

ch. 13 closes the cs-300 sort and list catalog. The lower bound of §13.0 is the floor that constrains every comparison sort — bubble (§13.1), quickselect's quicksort kin (§13.2), introsort, Timsort, dpqsort, pdqsort (§13.8). The non-comparison sorts (bucket §13.3, counting §13.6, radix §13.7) beat the floor by exploiting key structure. The list idioms (§13.4 container wrapper, §13.5 circular + sentinel, §13.10 cycle / reverse / merge / nth-from-end / intersection) fill out the algorithmic toolbox over the structure ch. 4 introduced. The decision table of §13.9 is the synthesis: every sort the reader will see in production code, indexed by the problem properties that select it.

Past the cs-300 boundary the catalog keeps extending: external sorting (multi-pass merge from disk for inputs that do not fit in RAM); GPU sorts (radix on the GPU is the dominant parallel sort); concurrent / lock-free sorts (parallel samplesort, ips4o). The production-references notebox above is the entry point.